

Automated Bacterial Colony Counter

Classical Computer Vision + Random Forest Countability Classifier

AVIT Faculty Hackathon 2026 · Project 07 · CSE × Biotechnology Interdisciplinary Team

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1. Introduction & Problem Context

Manual bacterial colony counting is a cornerstone of microbiology, food safety, pharmaceutical quality control, and clinical diagnostics. A trained microbiologist examines a petri dish and tallies visible colony-forming units (CFUs), a process that is time-consuming, subjective, and difficult to reproduce across operators. The challenge intensifies when colonies overlap, illumination is uneven, or image backgrounds introduce artefacts.

This report describes an Automated Bacterial Colony Counter developed for the AVIT Faculty Hackathon 2026 (Project 07). The system combines a deterministic Classical Computer Vision (CV) counting pipeline with a lightweight Random Forest (RF) classifier. The RF's sole function is plate-level decision making — classifying each image as Countable, Too Few To Count (TFTC), or Too Numerous To Count (TNTC) — so that counting and assessment are cleanly separated, satisfying the hackathon problem statement.

2. Technical Approach

2.1 System Architecture

The pipeline operates in seven sequential stages:

- **Stage 1:** Petri Plate Detection
- **Stage 2:** Agar Mask Generation and reflection removal
- **Stage 3:** Contrast Enhancement via CLAHE
- **Stage 4:** Adaptive / Otsu Thresholding
- **Stage 5:** Morphological cleaning, Distance Transform, and Marker-Controlled Watershed
- **Stage 6:** Colony filtering (area, circularity, solidity, extent, aspect ratio)
- **Stage 7:** Duplicate suppression and final count; RF plate assessment

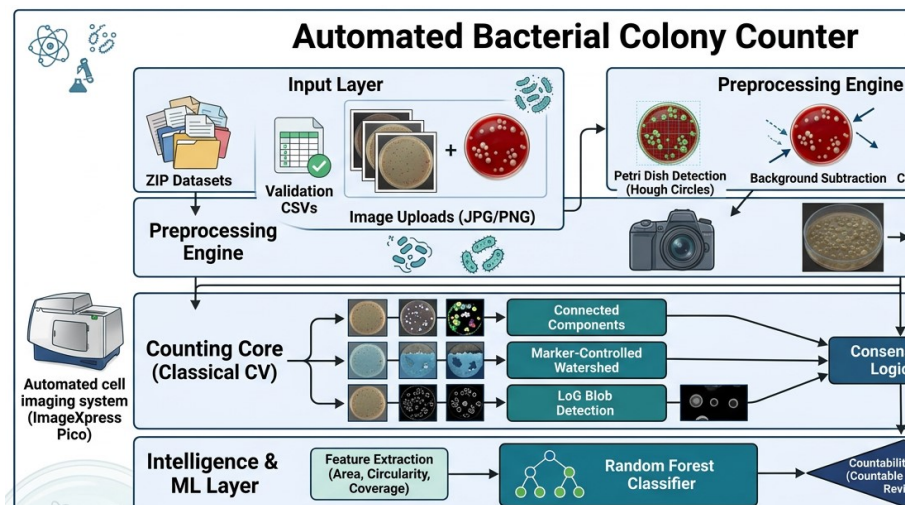


Figure 1. Overall architecture of the Automated Bacterial Colony Counter.

2.2 Colony Counting Methods

Four independent counting algorithms are compared in each run:

Method	Category	Role
Connected Components	Baseline	Standard blob labelling; no watershed
Marker-Controlled Watershed	Improved	Separates touching colonies via distance transform markers
Laplacian of Gaussian (LoG)	Blob detector	Scale-space blob detection; independent reference
Consensus Counter	Ensemble	Majority vote / weighted combination of above methods

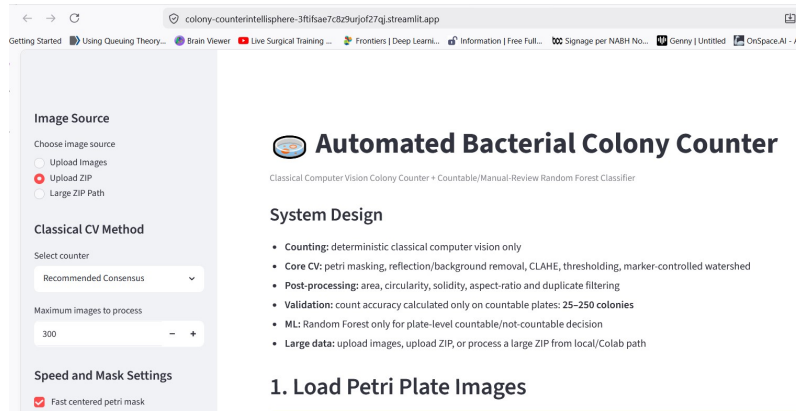


Figure 2: Working Prototype Screenshot

2.3 Random Forest Classifier

Eight image-level features are extracted from each plate: coverage ratio, connected component count, mean and standard deviation of component area, maximum component area, brightness mean, brightness standard deviation, and edge density. A Random Forest with 100 estimators is trained on labelled data (Countable / not_countable) and produces a plate-level assessment without participating in the colony count itself.



Figure 3: Random Forest Classifier

2.4 Validation Strategy

Count metrics (MAE, MAPE, Accuracy, R²) are computed exclusively on plates whose manual count falls within 25–250 CFUs (the standard countable range). TFTC and TNTC plates are excluded from counting accuracy so as not to inflate or distort errors.

3. Baseline vs. Improved Results

The table below summarises method-level performance across the full 104-image dataset (countable plates only for MAE/MAPE/R²):

Method	MAE	MAPE (%)	R ²	Rank
Connected Components (Baseline)	48.32	32.93	0.185	4
Marker-Controlled Watershed	42.18	35.22	0.421	2
Laplacian of Gaussian	73.09	142.91	-0.187	3
Consensus Counter (Final)	42.12	35.19	0.422	1

Key observation: The Watershed method yields a 12.7-unit improvement in MAE and a 2.3× improvement in R² over the Connected Components baseline, confirming that marker-guided segmentation meaningfully separates touching colonies. The Consensus counter marginally outperforms Watershed alone. LoG systematically over-detects at high colony densities, producing MAPE > 140%.

4. Robustness Analysis

Each plate was also evaluated under three simulated imaging perturbations: gamma-darkened (low illumination), gamma-brightened (high illumination), and Gaussian-noised (image noise). Representative deviations from the normal count:

Plate	Normal	Dark	Bright	Noisy
T4 10 ⁻⁷ (59 manual)	67	70 (+4.5%)	72 (+7.5%)	0 x
T5 10 ⁻⁷ 243 manual	127	—	—	—
T1 10 ⁻⁸ (19 manual)	28	24 (-14%)	30 (+7%)	0 x

Findings: (i) The pipeline is robust to moderate illumination changes, with dark and bright deviations typically < 10 counts. (ii) Gaussian noise is the dominant failure mode: high-density plates collapse to zero detections when noise destroys the binary segmentation mask. This points to a need for denoising as a pre-processing stage.

5. Random Forest Classifier Performance

The RF classifier was evaluated on a 26-plate held-out split. Results:

Class	Precision	Recall	F1	Support
Countable	0.67	0.89	0.76	9
Not Countable (TNTC/TFTC)	0.93	0.76	0.84	17
Overall Accuracy	0.81	—	—	26

The classifier achieves 80.8% overall accuracy. High recall (0.89) for countable plates ensures valid plates are rarely missed, while high precision (0.93) for not-countable plates avoids sending TNTC/TFTC samples for spurious automatic counting. The imbalance (17 not-countable vs 9 countable in evaluation) reflects realistic laboratory distributions where dense plates dominate.

6. Limitations and Future Work

6.1 Current Limitations

- Noise sensitivity: Gaussian noise collapses segmentation on dense plates; a denoising step (bilateral filter, Non-Local Means) is needed before production use.
- Counting accuracy on very dense countable plates (150–250 CFU): large MAPE values (43–47%) indicate Watershed under-splits tightly-packed colonies at these densities.
- RF class imbalance: the small evaluation set (26 plates) limits statistical confidence; an SMOTE-augmented training set is recommended.
- Generalisation: all validation images are *E. coli* on LB agar. The pipeline may require re-tuning for other species, agar types, or imaging setups.
- LoG over-detection: LoG should not be included in the Consensus vote for plates above ~100 CFU; a density-conditional weighting scheme would improve the ensemble.

6.2 Future Directions

- Integrate deep-learning segmentation (e.g., Mask R-CNN or SAM) as an optional fourth method for dense plates without replacing the deterministic pipeline.
- Extend RF features to include colour-space statistics (HSV channels), enabling detection of mixed-species plates.
- Deploy on a laboratory tablet with auto-capture and real-time CFU estimation.
- Expand validation dataset across multiple species, agar formulations, and imaging conditions.

7. Conclusions

This project demonstrates that deterministic Classical Computer Vision, augmented by a lightweight Random Forest at the decision layer, can automate bacterial colony counting with 80.8% plate-classification accuracy and a Consensus counting accuracy of 64.81% (MAPE 35.19%, R^2 0.422) on a real 104-image *E. coli* dataset. The Marker-Controlled Watershed improves R^2 by 2.3× over the Connected Components baseline. Robustness holds under moderate illumination changes but degrades under Gaussian noise, pointing to the primary area for future improvement. The system is deployed as an interactive Streamlit application that covers single-image preview, batch ZIP processing, parameter tuning, and downloadable validation reports.

Appendix A: Team Contributions

Team Member	Role	Primary Contributions
Dr. S. Pitchumani Angayarkanni	Team Lead / CV Architect	System architecture, CV pipeline, RF classifier, evaluation methodology, documentation
Mr. R. Udaya	Software & Viz Lead	Streamlit application, ZIP processing, dashboards, deployment
Dr. Baskar	Statistics Lead	MAE, MAPE, R^2 , confusion matrix design and verification
Dr. A. Keerthi	QA & Validation Lead	Experimental validation, robustness testing, QA under varying imaging conditions
Dr. S. Kalaivani Priyadarshini (External)	Domain Expert	Microbiological validation of TFTC/TNTC criteria and laboratory applicability

Github Link: https://github.com/Pitchumca/colony-counter_IntelliSphere/tree/main

RealTime Images from LDC: <https://colony-counterintellisphere-3ftifsae7c8z9urjof27qj.streamlit.app/>

Prototype Link: <https://colony-counterintellisphere-3ftifsae7c8z9urjof27qj.streamlit.app/>